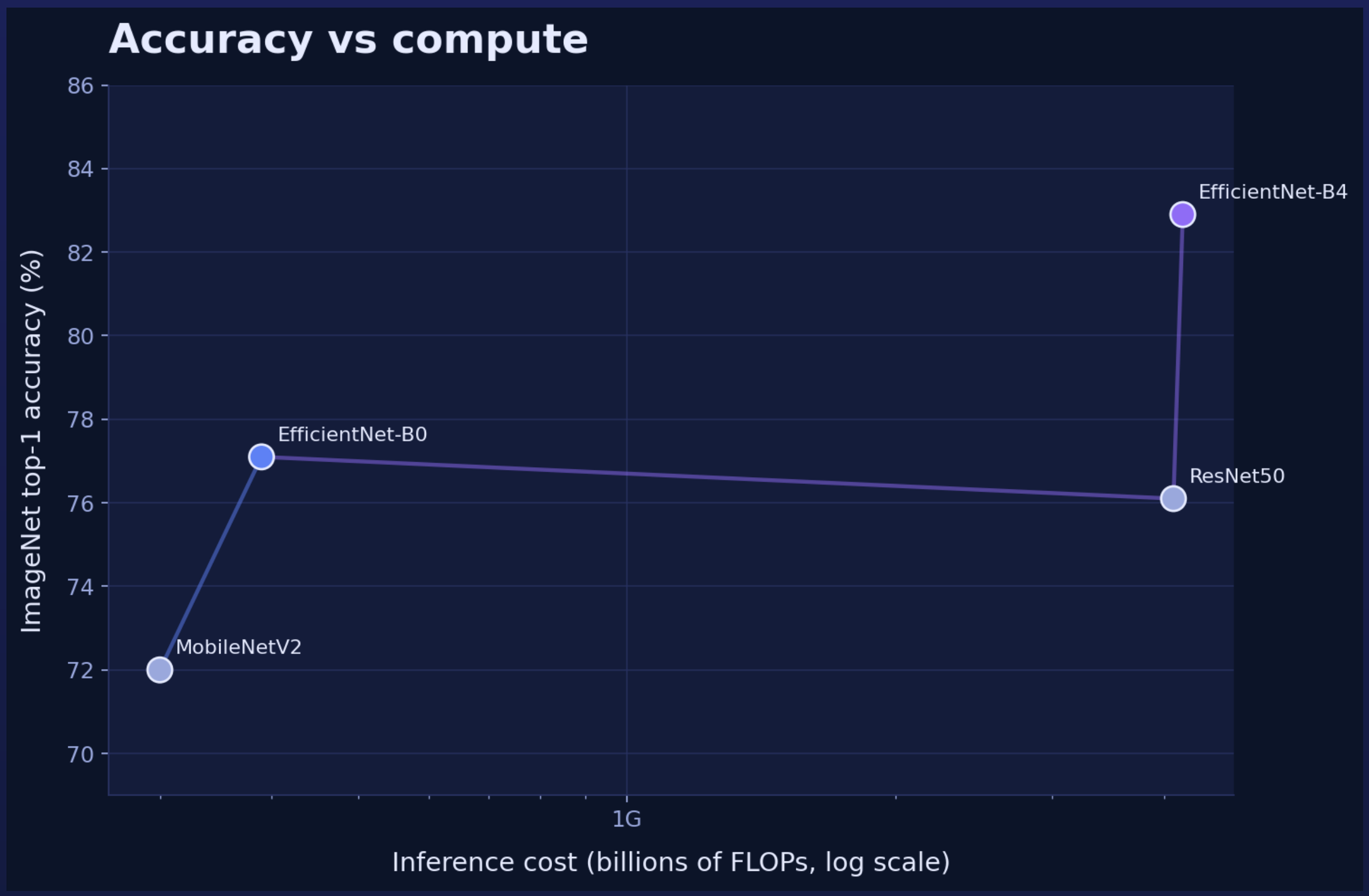
EfficientNet and the Problem With Scaling Models

ARTIFICIAL INTELLIGENCE · MACHINE LEARNING · DEEP LEARNING

EfficientNet and the Problem With Scaling Models

SWIPE ->

THE NUMBERS



EfficientNet-B0 beats ResNet50 accuracy at roughly one tenth of the compute.

When a model isn't accurate enough, a common instinct is to make it bigger.

Increase the depth.

Increase the width.

Increase the input resolution.

But scaling only one dimension doesn't always give the best balance between accuracy and computational cost.

EfficientNet introduced a compound scaling approach that scales three dimensions together:

- Network depth
- Network width
- Input resolution

The idea is to balance these dimensions instead of increasing one arbitrarily.

Why does this matter?

A deeper model can learn more complex feature hierarchies.

A wider model can represent more features at each layer.

Higher input resolution can preserve finer visual details.

Each has benefits, but each also has a cost.

EfficientNet uses a systematic scaling strategy to balance them.

For practical computer vision work, this is a useful reminder:

The best architecture isn't necessarily the largest one.

It's the one that achieves the right balance between predictive performance, latency, memory, and compute cost for your use case.

And that balance needs to be measured on the actual task, not assumed from a benchmark.

#EfficientNet #DeepLearning #MachineLearning

Thanks for reading.

Follow **Yasir Usman** for more on AI, ML and computer vision.

Code: github.com/yasirusman85/Tank